

## Module 4

# The Time Value of Money

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FINE 3010, Financial Management  
Prof. Renping Li, Tulane University

# Goals

- **Part 1**
  - Translate dollars today into dollars tomorrow (“future value”)
  - Translate dollars tomorrow into dollars today (“present value”)

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  - Financial calculators
- **Part 3**
  - Special cases: perpetuities, annuities, and their variants

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  - Translate dollars today into dollars tomorrow (“future value”)
  - Translate dollars tomorrow into dollars today (“present value”)
- **Part 2**
  - Financial calculators
- **Part 3**
  - Special cases: perpetuities, annuities, and their variants
- **Part 4**
  - The role of inflation

## Module 4, Part 1

# Future Value and Present Value

# The Time Value of Money



**Would you rather receive a dollar today, or a dollar tomorrow?**

# The Time Value of Money

- A dollar today is worth more than a dollar tomorrow



# The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?



# The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!



# The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!
  - Assuming, there is some investment with a positive return for us to pursue

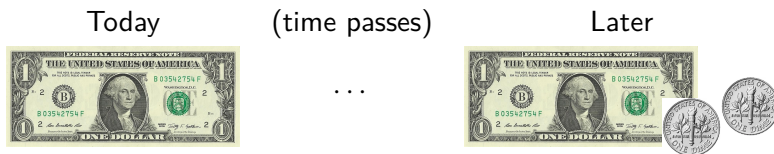


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$$\text{Future value (FV)} = \$100 \times (1 + 0.06)^5 = \$133.82$$

# Compound Interest

- **Compound interest**, the interest you earn with compounding, that is, with interest on interest

# Future Value, Compound Interest

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|------|---------------|-------------------------------|-------------|
| 1    | \$100.00      | $0.06 \times 100.00 = \$6.00$ | \$106.00    |
| 2    | \$106.00      | $0.06 \times 106.00 = \$6.36$ | \$112.36    |
| 3    | \$112.36      | $0.06 \times 112.36 = \$6.74$ | \$119.10    |
| 4    | \$119.10      | $0.06 \times 119.10 = \$7.15$ | \$126.25    |
| 5    | \$126.25      | $0.06 \times 126.25 = \$7.57$ | \$133.82    |

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- With compound interest, your \$100 grows to  $\$100(1 + r)^T = 133.82$ , so you earned \$33.82 in compound interest

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# Compound Interest

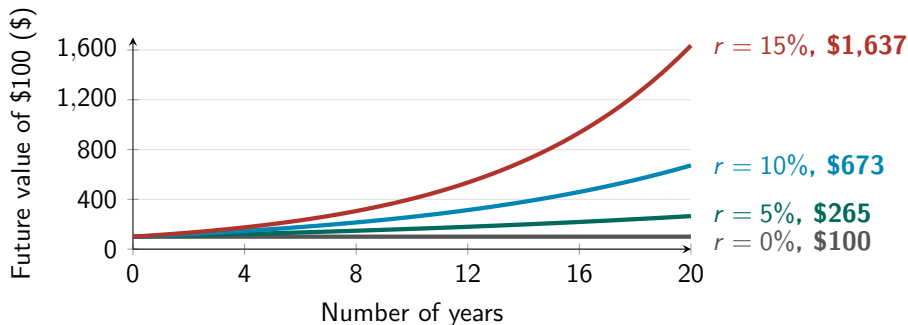
- **Compound interest**, the interest you earn with compounding, that is, with interest on interest
  - If you earn  $r$  per year for  $T$  years, your money grows to  $PV \times (1 + r)^T$
  - The total compound interest you have earned is  $PV \times (1 + r)^T - PV$

# Compound Interest

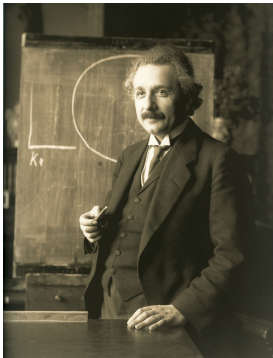
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- Compound interest is the default in finance (e.g., deposit rate)

## Future Value Grows Fast in $r$ and $T$

- The future value (FV) depends heavily on the interest rate  $r$ , especially as the time dimension  $T$  grows

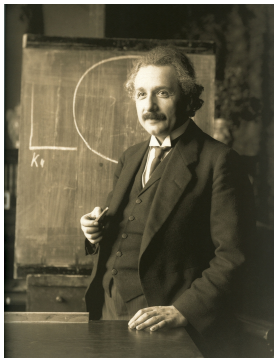


# What's the Most Powerful Force in the Universe?



- Einstein, ...

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- Einstein, ...
- “Compound interest”

## Example, Future Value

- Peter Minuit bought Manhattan Island for \$24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
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b.  $FV = \$24 \times (1 + 0.035)^{395} = \$19 \text{ million}$

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How many years will it take until your money doubles?



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  - We will learn how to calculate this with the financial calculator



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- \$100 in the future is worth less than \$100 in the present, so we “discount” the value of the future cash flow to its present value

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  - We choose a discount rate that reflects the risk in the cash flow, which we will return to in **Modules 8 and 9**

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- **Discount factor**, the present value of \$1 received in year  $T$  when the discount rate is  $r$ 
  - Less than one when  $r$  is positive

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 **Discount factor**

- **Discounted cash flow (DCF)**, the method of calculating present values by discounting future cash flows

## Example, Present Value

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

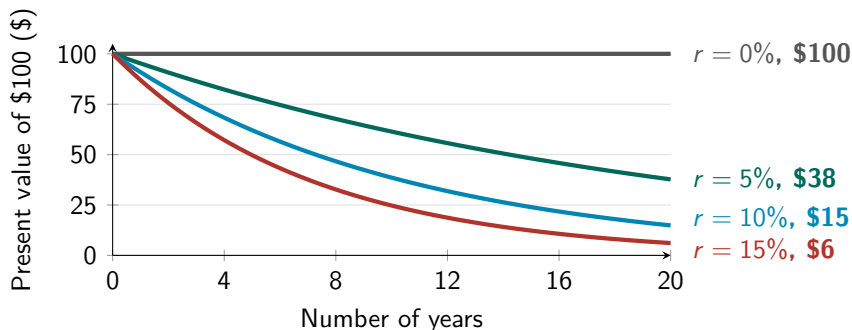
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$$\text{Present value (PV)} = \frac{\text{FV}}{(1 + r)^T} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

# Present Value

- The present value (PV) of \$100 in the future depends heavily on the discount rate  $r$ , especially as the time dimension  $T$  grows



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$$PV_{\text{total}} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots$$

- where  $C_1$  is the cash flow in one year,  $C_2$  is the cash flow in two years, etc.

## Example, Multiple Future Cash Flows

- You face two investments with different cash flows in the future
  - A. Receive \$15,000 cash now, or
  - B. Receive three payments, \$8,000 now, \$4,000 one year from now, and \$4,000 two years from now
- If the discount rate is 8%, which do you prefer?

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| 0                       | 1                                   | 2       | Year |
|-------------------------|-------------------------------------|---------|------|
| \$8,000                 | \$4,000                             | \$4,000 |      |
| ▼                       | ⋮                                   | ⋮       |      |
| \$8,000                 |                                     |         |      |
|                         | \$4,000/1.08 = \$3,704              |         |      |
|                         | \$4,000/1.08 <sup>2</sup> = \$3,429 |         |      |
| <hr/>                   |                                     |         |      |
| <b>Total = \$15,133</b> |                                     |         |      |

$$PV = \$8,000 + \frac{\$4,000}{(1 + 0.08)^1} + \frac{\$4,000}{(1 + 0.08)^2}$$

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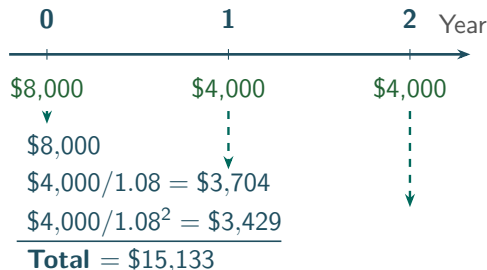
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- The present value of option B (\$15,133) is more than option A's \$15,000, so choose option B
- This is the principle behind the NPV rule in capital budgeting, which we will return to in **Modules 6 and 7**

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$$\text{Return} = \frac{\$180 - \$100}{\$100} = 80\%$$

# Holding Period Return

- **Holding period return**, percentage growth over the period an asset is owned, which can be longer or shorter than a year, e.g., owning a stock for  $T$  periods with returns  $r_1, r_2, \dots, r_T$  gives

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$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 1.80 \times 0.80 - 1 = 44\%$$

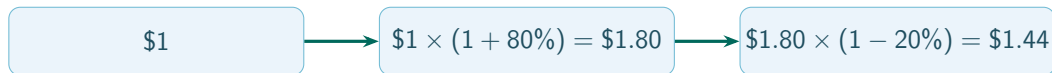
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- The annualized return is *not*  $\frac{80\% - 20\%}{2} = 30\%$

## Recap, Three Kinds of Returns

### Return

Percentage growth of an asset's value over some period of time

### Holding period return

Percentage growth over the whole period an asset is owned, longer or shorter than a year

### Annualized return

Percentage growth per year, a geometric mean over the  $T$  years of the holding period

## Recap, Three Kinds of Returns

| Return                                                         | Holding period return                                                                    | Annualized return                                                                     |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Percentage growth of an asset's value over some period of time | Percentage growth over the whole period an asset is owned, longer or shorter than a year | Percentage growth per year, a geometric mean over the $T$ years of the holding period |

- These financial returns are *not directly related in any way* with **Module 3's** accounting-based ROA and ROE measures

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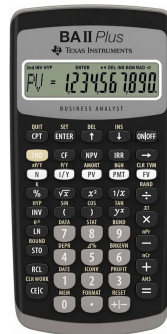
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## Module 4, Part 2

# Financial Calculators

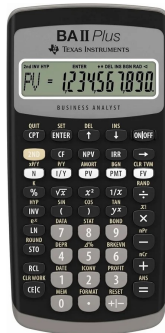
# Get the Texas Instruments BA II Plus

- You need a **financial calculator**



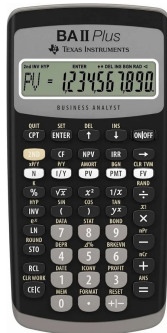
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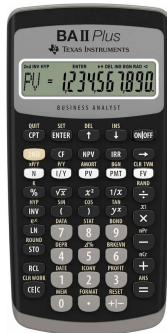
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- The CFA exams allow only two calculator series, and the **TI BA II Plus** is one of them



# Reset Your Financial Calculator

- If your financial calculator yields wrong answers, you may have changed some settings accidentally!

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- After you reset, you should repeat the next two settings!

# Set the Number of Digits to at Least 4!

- On the TI BA II Plus, press
  - **2nd**
  - **.**
  - **4**
  - **ENTER**

# Set the Number of Compounding Periods per Year to 1!

- On the TI BA II Plus, press
  - **2nd**
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  - 1
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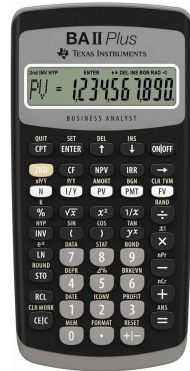
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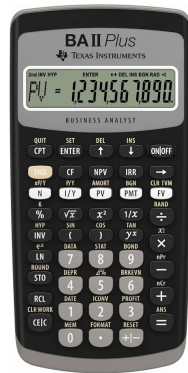
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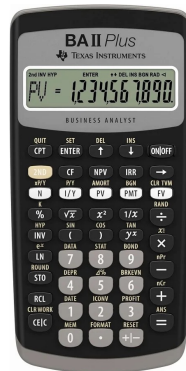
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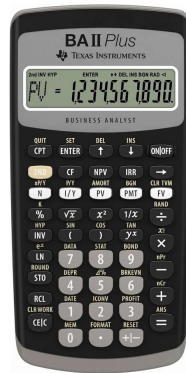
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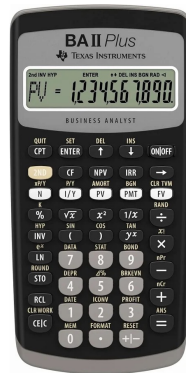
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- Why does the answer come out negative?

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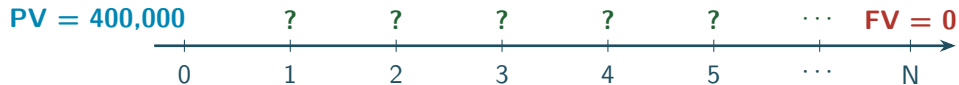
**The present values of PV, all PMTs, and FV, summed up together equal 0.**

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- That is, you would pay an undiscounted total of  $30 \times \$25,430$  over the course of your mortgage

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- Set the number of compounding periods per year to 1, and show at least 4 digits

## Module 4, Part 3

# Special Cases: Perpetuities and Annuities

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  - Neat shortcuts make it very easy to calculate the present value of these patterns of cash flows

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- A perpetuity has a present value of

$$\text{PV of perpetuity} = \frac{\text{Cash flow}}{\text{Discount rate}} = \frac{C}{r}$$

## Example, Perpetuities

- Suppose the San Antonio Spurs offer Victor Wembanyama a perpetual payment of \$20m every year. He will be paid forever, as long as he lives, and then his children, and then his children's children, etc.
- Suppose the discount rate is 10%. What is the present value of this contract?



## Example, Perpetuities

- Solution,

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$$PV = \frac{C}{r} = \frac{20}{10\%} = \$200m$$

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- **Delayed perpetuity**, first payment at time  $t > 1$

$$PV = \frac{C}{r} \times \frac{1}{(1+r)^{t-1}}$$



## Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at  $C$  but grows at a constant growth rate  $g$  forever. This is a growing perpetuity

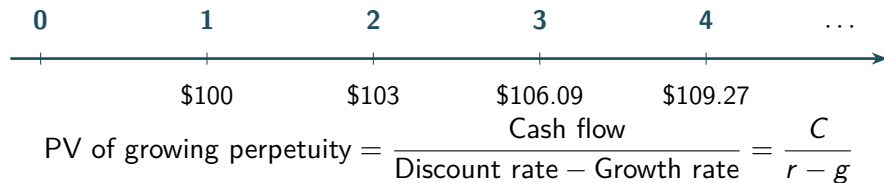
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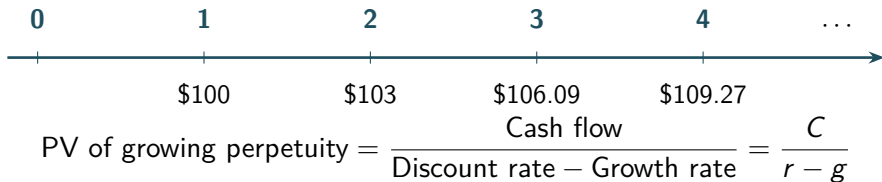
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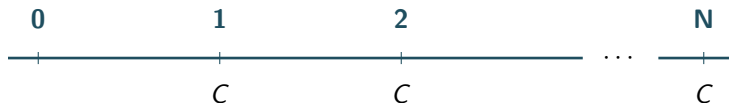
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- This formula comes up often when valuing stocks, e.g., in the “Gordon growth model” and DCF valuation (more in [Module 5](#))
- When  $g = 0$ , the PV simplifies to the formula for a regular (non-growing) perpetuity

# Annuities

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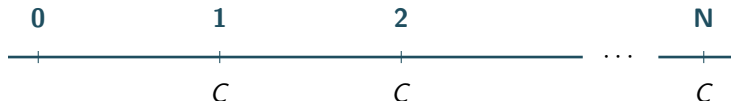


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- $C$  is the annual cash flow (aka the payment),  $r$  is the discount rate, and  $N$  is the number of years that the annuity lasts

## Example, Annuity

- You are purchasing a car. You are scheduled to make 3 annual installments of \$8,000 per year, with the first payment a year from now. Given a discount rate of 10%, what is the price you are paying for the car (i.e., what is the PV of the annuity)?

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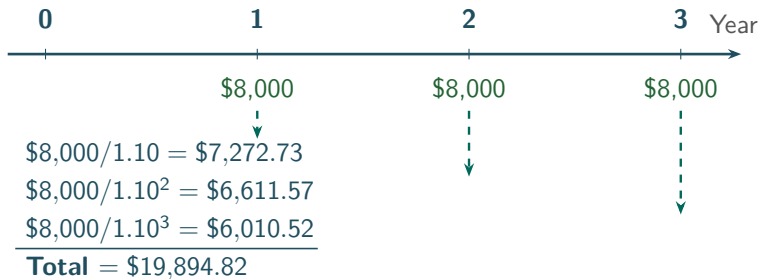
$$PV = C \frac{1}{r} \left( 1 - \frac{1}{(1+r)^N} \right) = \$8,000 \times \frac{1}{0.10} \times \left( 1 - \frac{1}{(1+0.10)^3} \right) = \$19,895$$

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- What about a back-loaded cash flow and an annuity?

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## Practice Problem

- A store sells a product for \$100. It offers you two payment plans,
  - Under the installment plan, you pay \$25 today and \$25 at the end of each of the next 3 years
  - If you pay the entire bill today, you get a 7% discount and pay only \$93
- a. Suppose the interest rate is 4%. What is the present value of your payments under the installment plan?
- b. Which plan is a better deal?

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- b. Paying in full now is better (\$93 beats \$94.38)
- On a financial calculator, **PMT** = -25, **N** = 3, **FV** = 0, **I/Y** = 4, leading to 69.38, and next,  $69.38 + 25 = 94.38$

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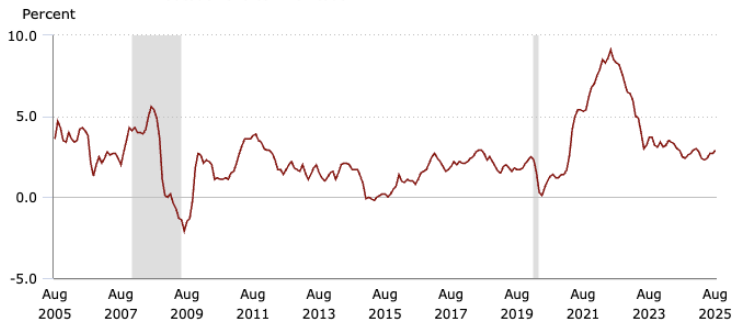
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- These have handy formulas for calculating their present values!
- Timing,
  - Ordinary perpetuities and annuities make their first payment at time  $t = 1$
  - Perpetuity and annuity dues make their first payment at  $t = 0$
  - Delayed perpetuities and annuities make their first payment at  $t > 1$

## Module 4, Part 4

# Inflation

# History of Inflation



Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.  
Source: U.S. Bureau of Labor Statistics.

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- **Nominal interest rate or nominal return**, the rate at which money invested grows
  - Compare dollars at time  $t = 0$  to dollars at time  $t = 1$
- **Real interest rate or real return**, the rate at which purchasing power increases
  - Compare the ability to buy a basket of goods at time  $t = 0$  to the ability to buy the same basket of goods at time  $t = 1$

# Inflation and Returns

- In the presence of inflation, an investor's real return (e.g., the real interest rate) is always less than the nominal return

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- Inflation also affects discounting,
  - We should discount nominal dollar cash flows using a nominal discount rate
  - We should discount real dollar cash flows using a real discount rate

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- Approximation,  $\text{real} \approx 6\% - 2\% = 4\%$

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- $(1 + \text{Real return}) = (1 + \text{Nominal return}) / (1 + \text{Inflation rate})$ , and approximately,  $\text{real return} \approx \text{nominal return} - \text{inflation rate}$
- Discount nominal cash flows using a nominal discount rate, and real cash flows using a real discount rate

# Key Takeaways

- A dollar today is worth more than a dollar tomorrow,  $FV = PV \times (1 + r)^T$  and  $PV = FV / (1 + r)^T$  translate between them

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- A dollar today is worth more than a dollar tomorrow,  $FV = PV \times (1 + r)^T$  and  $PV = FV / (1 + r)^T$  translate between them
- Perpetuities and annuities have shortcut PV formulas, mind the timing of the first payment
- The real return strips inflation out of the nominal return

# Thank you!

Please let me know if you have any questions.